Date: 02 July 2026	**Team ID:** SB-7429
Project Name: Credit Card Approval Prediction using Machine Learning	**Maximum Marks:** 3 Marks

PROJECT EXECUTABLE FILES TEMPLATE

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (detailed instructions to run)	Yes
3	requirements.txt (system dependencies configuration file)	Yes
4	Fitted scaler weights binary (scaler.pkl)	Yes
5	Calibrated rejection boundary float (threshold.pkl)	Yes
6	Trained Random Forest Classifier binary (credit_model.pkl)	Yes
7	Dataset cache directories (application_record.csv & credit_record.csv)	Yes
8	Unit and integration test suite scripts (test_app.py)	Yes
9	Static EDA visualization assets (income/education distributions)	Yes
10	Hosted public deployment URL link	Yes

Step 2: File / Folder Structure

The layout of the project workspace is configured as follows:

```
creditcard_approval_project/
■■■ 1. Brainstorming & Ideation/ (Brainstorming_and_Ideation.md)
■■■ 2. Requirement Analysis/ (Requirements_Specification.md)
■■■ 3. Project Design Phase/ (System_Design.md)
■■■ 4. Project Planning Phase/ (Project_Plan.md)
■■■ 5. Project Development Phase/ (Development_Log.md)
■■■ 6.Project Testing/ (Testing_Strategy.md, test_app.py)
■■■ 7.Project Documentation/ (User_and_API_Manual.md)
■■■ 8.Project Demonstration/ (Demonstration_Guide.md)
■■■ static/
■ ■■■ css/ (style.css - dark theme styling)
■ ■■■ images/ (static EDA charts)
■■■ templates/
■ ■■■ home.html (applicant attributes input form)
■ ■■■ result.html (decision outcome dashboard)
■■■ app.py (Flask web controller, policy override filters)
■■■ model.py (training engine, SMOTE balancing, threshold calibration)
■■■ eda.py (EDA graphics generator)
■■■ requirements.txt (packages list)
■■■ Procfile (production deployment configurations)
■■■ credit_model.pkl (Random Forest binary)
■■■ scaler.pkl (StandardScaler binary)
■■■ threshold.pkl (calibrated float threshold)
```

Step 3: Deployment / Access Details

Field Name	Access / Configuration Details
Hosted / Deployed URL	https://credit-card-approval-v2.onrender.com
Login Credentials (Demo)	Public Web Service (No authorization walls required for demo).
Platform / Hosting Provider	Render Cloud Platform (Free Web Service Tier).
Repository Link	https://github.com/MaheshKona7/credit-card-approval-v2
Demo Video Link	https://youtu.be/example_credit_card_demo

Step 4: Run Instructions

To set up and execute the application locally:

1. Open the project root terminal and initialize the environment:
   python -m venv venv
   .\venv\Scripts\Activate.ps1 (Windows)
2. Install all required dependencies:
   pip install -r requirements.txt
3. Run the model pipeline to train the model and generate serialized binaries:
   python model.py
4. Start the Flask server web client interface:
   python app.py
5. Open your web browser and navigate to: **http://127.0.0.1:5000/**

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Current Status
1	Stratified merge is slow when loading raw tables from disk.	Solved: Pre-processed datasets are cached locally to speed up subsequent server executions.
2	Computer policy blocks C-DLL execution of XGBoost on local machines.	Resolved: Bypassed XGBoost; utilized scikit-learn's Random Forest which runs natively.